

What is TypeScript? 🤔

TypeScript is a **superset of JavaScript Programming Language** developed by [Microsoft](#) that adds **static typing**, other useful features to make code safer and easier to maintain.

Or

TypeScript is an open-source programming language developed by Microsoft that extends JavaScript by adding static typing and advanced development features, helping developers build safer, more maintainable, and scalable applications.

In simple words:

TypeScript = JavaScript + Extra Features + Type Safety

What Does "Superset of JavaScript" Mean?



A superset means it includes everything that JavaScript has, plus additional features.

For example:

```
let name = "John";  
console.log(name);
```

This JavaScript code is also valid TypeScript code.

Every JavaScript program can run in TypeScript.

What is Static Typing? 🤔

Static typing means TypeScript checks data types while writing code, before the application runs.

Example:

```
let age: number = 25;
```

Here TypeScript knows:

```
age → number
```

So this is valid:

```
age = 30;
```

But this is invalid:

```
age = "Thirty";
```

TypeScript shows an error before execution.

Why TypeScript Was Created?

JavaScript is dynamically typed.

```
let age = 25;  
age = "Twenty Five";  
age = true;
```

JavaScript allows this.

The problem is:

- Bugs are found at runtime

- Large projects become difficult to maintain
- Refactoring becomes risky
- Team collaboration becomes harder

TypeScript catches such errors during development.

```
let age: number = 25;  
  
age = "Twenty Five"; // Error
```

TypeScript prevents bugs before code reaches production.

Example 1

Without TypeScript

```
function add(a, b) {  
  return a + b;  
}  
  
add(10, "20");  
Output:  
1020
```

Unexpected behavior.

With TypeScript

```
function add(a: number, b: number) {  
  return a + b;  
}  
  
add(10, "20");
```

Error appears before running.

Example 2

```
const user = {  
  name: "Dev",  
  age: 25  
};  
  
user.age = "twenty five";
```

Error:

Type 'string' is not assignable to type 'number'